

Scalpel and Code: Integrating Artificial Intelligence for Comprehensive Surgical Performance Evaluation

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Presentation date: 09/18/2026

- ***Nothing to Disclose***

BACKGROUND

The Assessment Gap in Surgical Training

Training Readiness Gap

National data describe persistent gaps in chief-resident preparedness and declining operative autonomy. [1,2]

Skill Affects Outcomes

Technical skill has been associated with complication rates and patient outcomes. [3]

Assessment burden

OSATS and video review improve structure but remain time-consuming and vulnerable to rater variability and bias. [4,7-9]

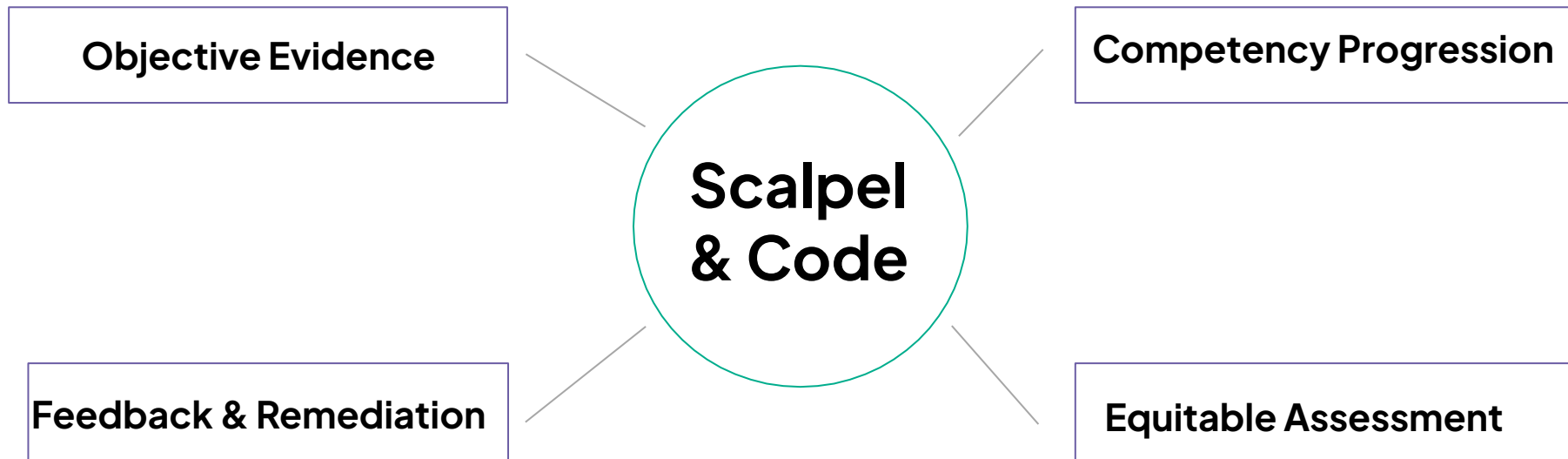
Video +AI Opportunity

Video based assessment and machine learning can support standardized, data-driven feedback while reducing manual review burden. [5,6,10-12]

PROJECT RATIONALE

A Data-Driven Assessment Engine for Surgical Performance

AI-enabled video analysis can convert operative video into objective motion and workflow data.



PROJECT OBJECTIVES

Aim 1 – Objective Assessment

Can an AI-powered motion tracking system objectively assess technical surgical skill?

Aim 2 – Discrimination

Can motion analytics distinguish between different levels of surgical experience?

Aim 3 – Validation

Do AI-generated metrics correlate with expert ratings and predict performance improvement over time?

Prototype

Trained detector + tracker

Measurement

Motion + idle + coordination

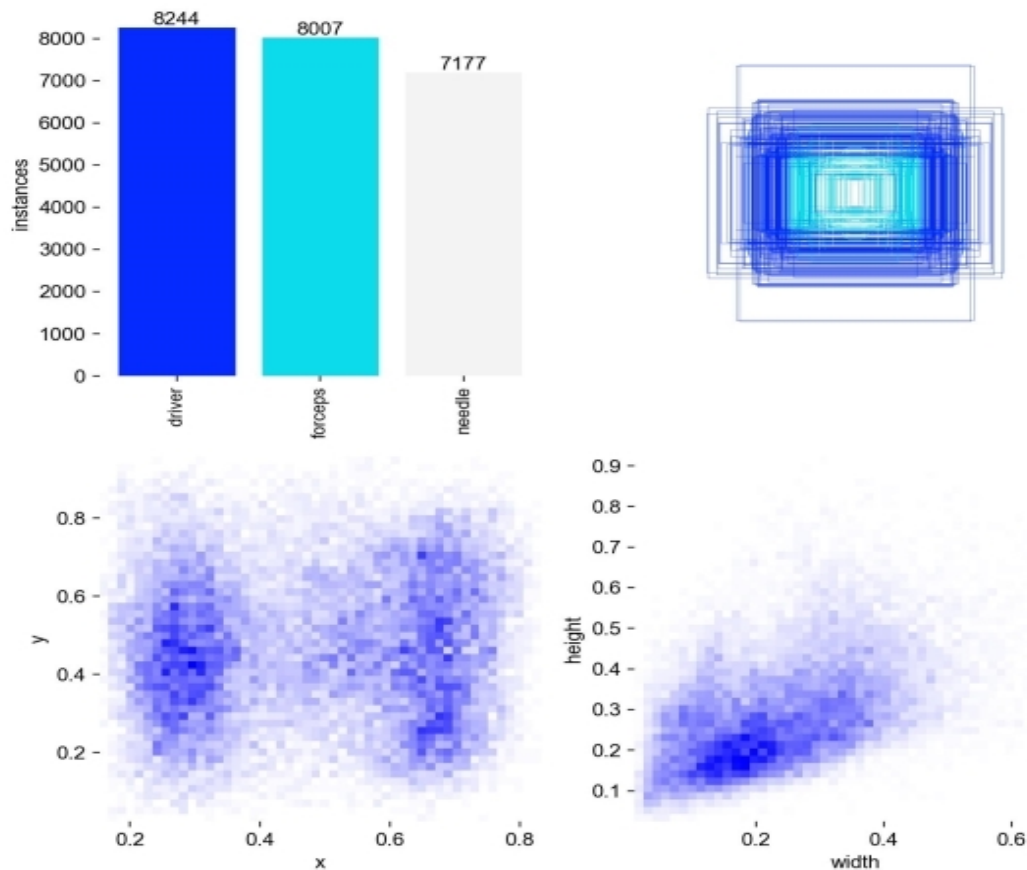
Quality Control

PASS/CAUTION/FAI
L gating

Planned Validation

Expert Rating
Correlation

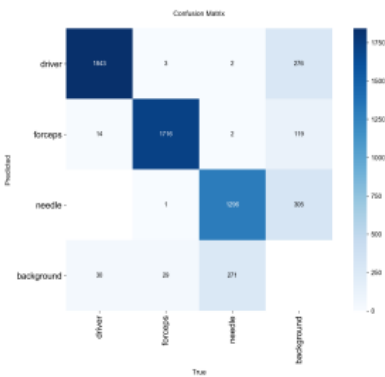
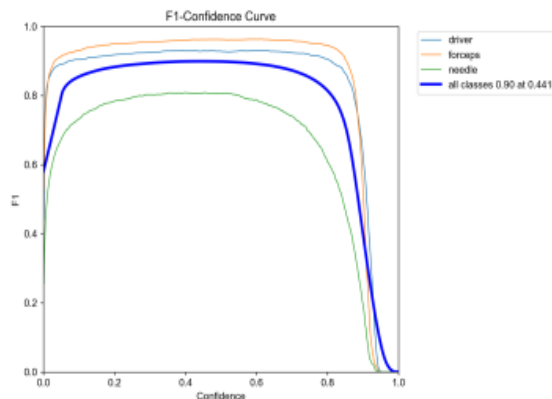
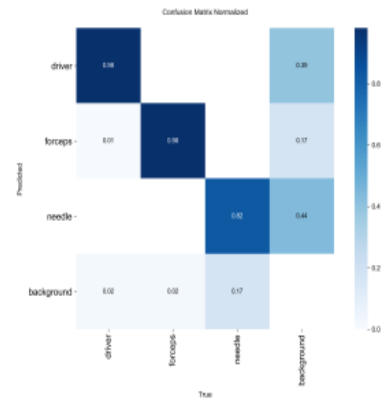
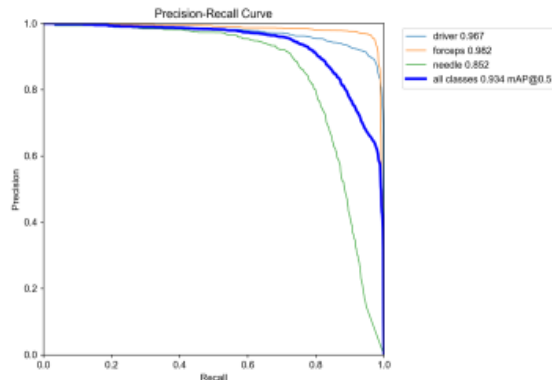
Retrospective Quantitative Pipeline



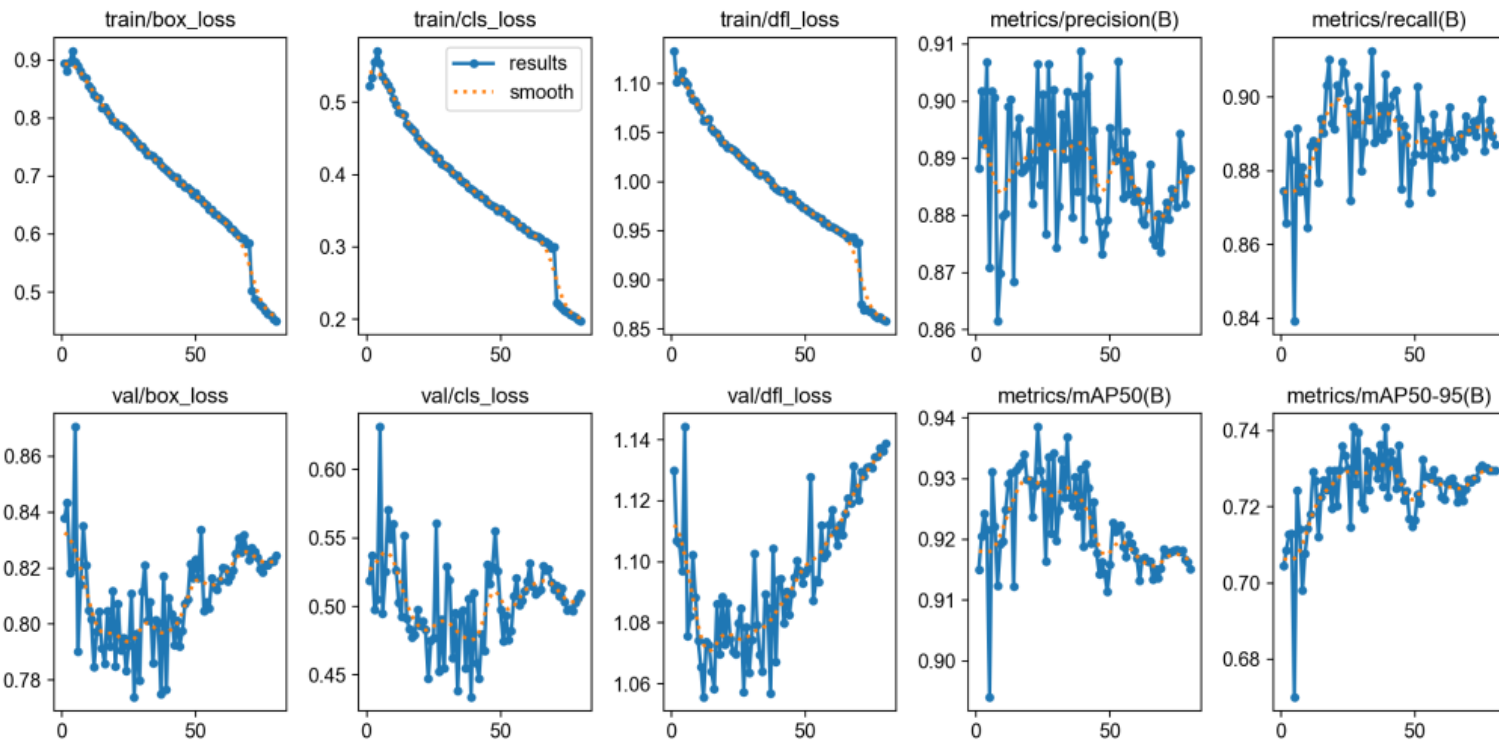
Detector Performance

Performance supports downstream tracking-readiness, not clinical validation

Metric	Results
Overall precision	0.906
Overall recall	0.896
Overall mAP@0.5	0.934
Overall mAP@0.5–0.95	0.741
Driver AP@0.5	0.967
Forceps AP@0.5	0.982
Needle AP@0.5	0.852



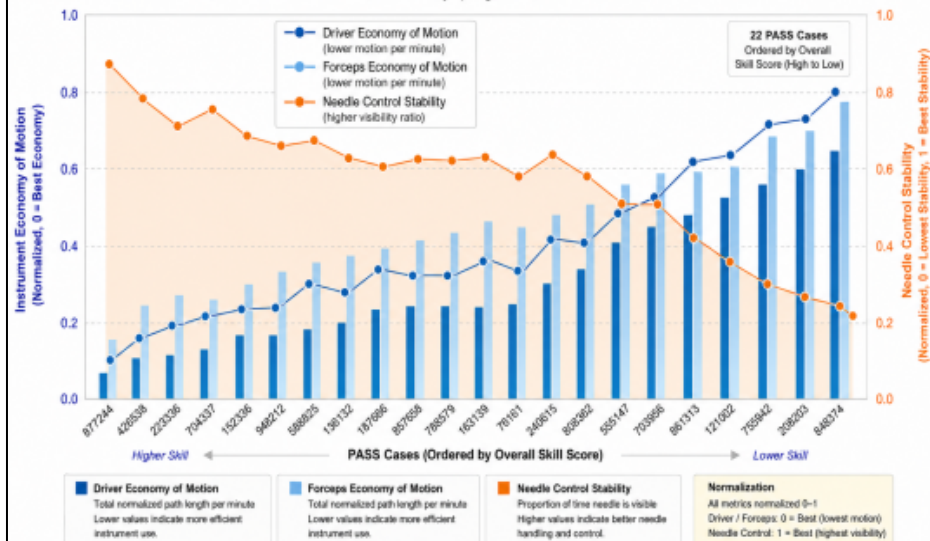
Current Interpretation of Results



Provisional AI-Derived Performance Patterns Across QC Passed Cases

Integrated Instrument Economy and Needle Control Across PASS Cases

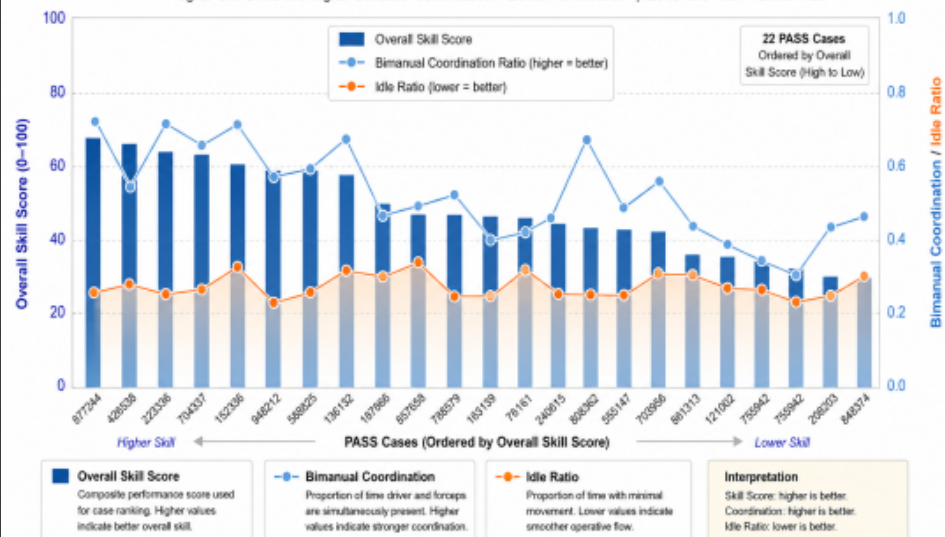
Lower Instrument Motion = Better Economy | Higher Needle Control = Better Performance



This integrated view shows that higher skill cases achieve: lower instrument motion (better economy) and higher needle control stability.

Integrated Skill Score, Bimanual Coordination, and Idle Ratio Across PASS Cases

Higher Skill Score and Higher Bimanual Coordination = Better Performance | Lower Idle Ratio = Better Flow



This integrated view shows that higher skill cases generally demonstrate stronger bimanual coordination and lower idle time.

FUTURE DIRECTIONS

**Phase 2
testing**

**QC
comparison**

**Expert
ratings**

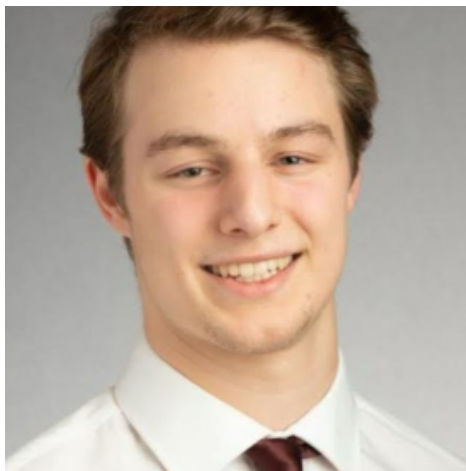
**Validity
Analysis**

**Model
Refinement**

**Multi-site
Expansion**



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THANK YOU

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